

number theory part 1-exam

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score
Question 1	3 / 3
Question 2	3 / 3
Question 3	1 / 1
Question 4	1 / 1
Question 5	1 / 1
Question 6	1 / 1
Question 7	1 / 1
Question 8	1 / 1
Total	12 / 12 (100%)

Performance Summary

Exam Name:	number theory part 1-exam
Session ID:	03270705203
Exam Start:	Sun Jul 12 2026 20:45:59
Exam Stop:	Sun Jul 12 2026 21:52:36
Time Spent:	1:06:36

Question 1

Bézout Coefficients

Find the Bézout coefficients and the gcd of 88 and 67:

16

Expected answer: 16 $\times 88 +$

-21

Expected answer: -21 $\times 67 =$

1

Expected answer: 1**a**Your answer is correct. You were awarded **1** mark. **b**Your answer is correct. You were awarded **1** mark. **gcd**Your answer is correct. You were awarded **1** mark. You scored **3** marks for this part.

Score: 3/3

Question 2

Bézout Coefficients

Find the Bézout coefficients and the gcd of 47 and 87:

-37

Expected answer: -37 $\times 47 +$

20

Expected answer: 20 $\times 87 =$

1

Expected answer: 1**a**Your answer is correct. You were awarded **1** mark. **b**

Your answer is correct. You were awarded **1** mark. ✓

gcd

Your answer is correct. You were awarded **1** mark. ✓

You scored **3** marks for this part.

Score: 3/3 ✓

Question 3

Modular Inverse

What is the modular inverse of $89 \bmod 28$? (Make sure the number is between 0 and 27).

17 ✓

Expected answer: 17

Your answer is correct. You were awarded **1** mark. ✓

You scored **1** mark for this part.

Score: 1/1 ✓

Question 4

Modular Inverse

What is the modular inverse of $79 \bmod 91$? (Make sure the number is between 0 and 90).

53 ✓

Expected answer: 53

Your answer is correct. You were awarded **1** mark. ✓

You scored **1** mark for this part.

Score: 1/1 ✓

Question 5

Find the solution to the following equations:

$$x = 0 \pmod{5}$$

$$x = 10 \pmod{19}$$

10



Expected answer: 10

Your answer is correct. You were awarded **1** mark. ✓You scored **1** mark for this part.

Score: 1/1 ✓

Question 6

Find the solution to the following equations:

$$x = 0 \pmod{2}$$

$$x = 26 \pmod{29}$$

26



Expected answer: 26

Your answer is correct. You were awarded **1** mark. ✓You scored **1** mark for this part.

Score: 1/1 ✓

Question 7

Modular Inverse

Solve for x in $70x \equiv 41 \pmod{61}$? so that x is between 0 and 60.



Expected answer: 52

Your answer is correct. You were awarded 1 mark. ✓

You scored 1 mark for this part.

Score: 1/1 ✓

Question 8

Modular Inverse

Solve for x in $13x \equiv 4 \pmod{8}$? so that x is between 0 and 7.



Expected answer: 4

Your answer is correct. You were awarded 1 mark. ✓

You scored 1 mark for this part.

Score: 1/1 ✓

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